**DATE:** October 15, 2026

**TO:** Dr. Haris Vikalo
Mugdha Jadao

**FROM:** Ethan Chua
Esteban Martinez
Li Shi
Andrew Yin
Abhirit Das

**SUBJECT:** Project Definition for BevoTag, an Open Asset-Tracking Tag for the UT Austin Campus

## INTRODUCTION

This Project Definition documents our team's understanding of the BevoTag design problem and describes how the team has organized itself to solve it. BevoTag is a small tracking tag that a student or staff member can attach to a laptop, bag, or piece of university equipment to locate it anywhere on the University of Texas at Austin campus. Consumer tags such as the Apple AirTag locate an item only when a nearby phone from the same manufacturer happens to detect it [1], and satellite positioning such as GPS fails inside buildings. BevoTag instead determines its own position from the Wi-Fi networks already installed across campus and from a built-in motion sensor, then reports that position to a map-based application.

The report first explains the problem in plain terms: why existing trackers and satellite positioning fall short on campus, how combining Wi-Fi signal measurements with motion sensing can provide a continuous location estimate, what the tag will and will not do, and the standards and ethical obligations, particularly location privacy, that shape any solution. It then translates these needs into measurable requirements, including a median position error under 5 m and correct room identification in more than 90% of trials within surveyed buildings. Finally, the report assigns the five team members to a hardware and firmware sub-team and a localization and software sub-team under a single project lead, and it sets the policies the team will use to communicate, track work, and resolve disagreements.

## PROBLEM STATEMENT

Locating a small object reliably across a dense urban campus is difficult because radio signals reflect off and are blocked by buildings, and low-cost motion sensors drift over time [2]. This section first describes the design problem, including the need that motivates BevoTag, the technical approach it relies on, its functions and scope, and the standards and ethical issues that constrain it. It then defines the inputs, outputs, interfaces, operating conditions, and performance targets that a

successful design must meet.

## Design Problem Description

A successful BevoTag design must estimate the location of a tagged item indoors and outdoors, without relying on infrastructure the team cannot control, and must present that location to its owner in a form that is easy to interpret. The following subsections explain why this need exists, why current products and satellite positioning do not meet it, the principle on which BevoTag operates, and the scope, challenges, standards, ethical considerations, and deliverables that follow from it.

### *The Need for On-Campus Item Tracking*

Students, faculty, and staff routinely lose valuable items across the 40 Acres. Laptops, bags, and shared university equipment move between classrooms, labs, libraries, and outdoor plazas many times a day, and an item left behind can be anywhere among dozens of multi-story buildings. A tracking tag on campus must therefore work indoors, on any floor, and in the narrow outdoor spaces between tall buildings.

BevoTag serves UT Austin students, faculty, and employees who need to track personal belongings or university equipment across campus. The team's motivation is to provide tracking that requires neither new infrastructure, such as beacons installed in every building, nor a passerby who owns a particular brand of smartphone.

### *Why Satellite Positioning Fails on Campus*

Most outdoor tracking relies on Global Navigation Satellite Systems (GNSS), the family of satellite positioning systems that includes GPS. A GNSS receiver computes its position from faint signals broadcast by satellites more than 20,000 km away. Walls and roofs block those signals almost entirely, so GNSS stops working inside buildings. Between tall buildings, the signals that do arrive often reach the receiver only after reflecting off nearby walls, a phenomenon called multipath, which corrupts the position estimate by tens of meters [2]. GNSS receivers also consume considerable power while searching for satellites, which shortens battery life in a small tag.

### *Limitations of Existing Tracking Solutions*

Current products avoid some of these problems but introduce others. Crowdsourced tags such as the Apple AirTag and Samsung SmartTag broadcast a short-range Bluetooth signal and depend on nearby phones from the same manufacturer to detect that signal and report its location [1]. When no such phone passes by, as in an empty lab at night, the tag's location stops updating, and

the system offers no access to its data for other applications. Dedicated GNSS trackers suffer the indoor signal loss and power drain described above.

A third approach estimates position from the received signal strength indicator (RSSI), a measure of how strong each nearby Wi-Fi signal is when it reaches the device. Because signal strength changes as people walk by, doors open, and signals reflect off walls, RSSI-only systems commonly mislocate a device by 5 to 15 m [3], which is often enough to place it in the wrong room.

***How BevoTag Determines Its Location***

BevoTag combines two complementary sources of information, neither of which is sufficient on its own. The first is the campus Wi-Fi network. UT Austin operates Wi-Fi access points throughout its buildings and plazas, including the `utexas` and `utexas-iot` networks [4]. In addition to RSSI, some Wi-Fi chips can report channel state information (CSI), which describes how a signal was weakened and delayed on each of dozens of narrow frequency sub-channels. Because walls, furniture, and room geometry shape these values, CSI acts as a detailed fingerprint of a location. By first surveying a building, which means recording Wi-Fi measurements at many points whose positions are known, the system can later estimate the tag's position by matching its current measurements to the survey.

The second source is an inertial measurement unit (IMU), a motion sensor similar to the one that rotates a smartphone's screen. A six-degree-of-freedom IMU measures acceleration along and rotation about three axes. Adding up these measurements from a known starting point, a technique called dead reckoning, tracks the tag's movement many times per second, but small sensor errors accumulate, so the estimate drifts further from the truth the longer it runs [5]. An extended Kalman filter (EKF), a standard algorithm for combining noisy measurements, merges the two sources: it uses the IMU to predict how the tag has moved and uses each new Wi-Fi position fix to correct the accumulated drift, weighting each source by how reliable it currently is. The result is a position estimate that is smoother than Wi-Fi alone and does not drift like the IMU alone.

***Design Functionality and Scope***

Figure 1 shows how the parts of the BevoTag system work together. The tag measures nearby Wi-Fi signals and its own motion (step 1) and sends these measurements over the campus Wi-Fi network (step 2) to a server. The server compares the Wi-Fi measurements with a previously recorded survey of the building and fuses them with the motion data to estimate the tag's position (step 3). The user application then displays that position on a campus map (step 4).

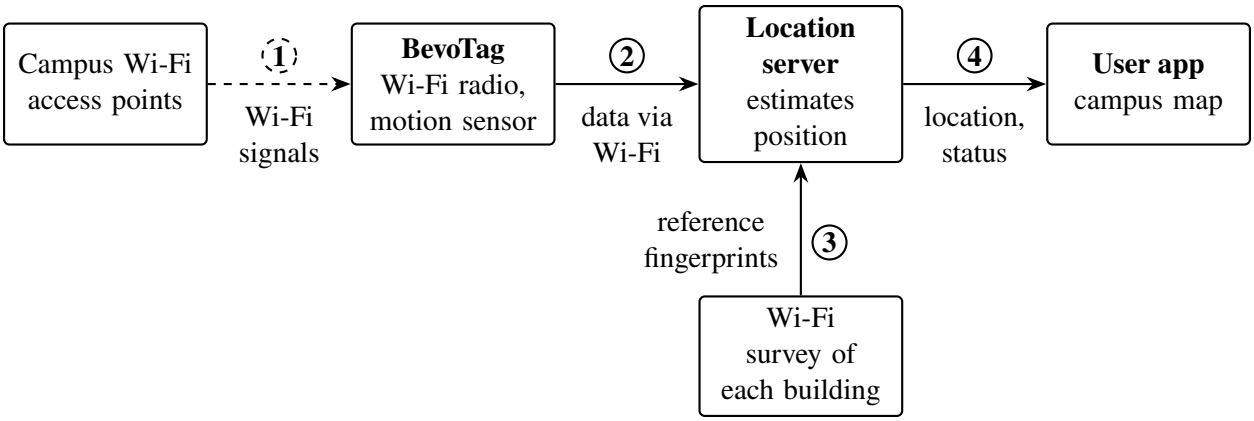



**Figure 1. How BevoTag Locates a Tagged Item**

To deliver this behavior, any BevoTag design must perform four functions:

- **Wi-Fi measurement:** record the RSSI and, where the hardware supports it, the CSI of signals from nearby campus access points in the 2.4 GHz band.
- **Motion tracking:** sample the IMU's acceleration and rotation fast enough to follow a walking person carrying the tag.
- **Position estimation:** fuse the motion data with Wi-Fi position fixes, correcting the drift that accumulates in the motion estimate.
- **Reporting:** send the latest position and device status to an application that the owner can view.

The project includes a working tag built on a developer board or custom circuit board that combines a Wi-Fi chip able to capture CSI, such as the ESP32-S3, with a low-noise IMU. It also includes the firmware that reads the sensors and runs or supports the position estimate, a survey of Wi-Fi measurements in designated test areas beginning with the Engineering Education and Research Center (EER) and its adjacent outdoor plazas, and a web or mobile application that displays the tag's position on a campus map.

Several related capabilities fall outside the project's scope. The tag will not use a cellular connection such as LTE-M or NB-IoT; these require a paid data plan and a second radio that would increase cost and power consumption, and campus Wi-Fi already covers the target areas. The team will not survey every building on campus, because each survey requires hours of walking measurements per floor; instead, testing will be limited to a few defined benchmark areas, and the survey method will be documented so that additional buildings can be added later. The prototype enclosure will be 3D-printed or purchased rather than injection-molded for mass production,

since the project's goal is a working proof of concept rather than a manufacturable product. Finally, BevoTag is designed to locate objects, not people, and features intended to follow a person, such as continuous covert tracking, are excluded by design.

***Design Challenges***

Three challenges will most strongly shape the design. The first is the trade-off between computing on the tag and transmitting raw data. Processing CSI and running the EKF on a small microcontroller strains its memory and battery, whereas sending raw measurements to the server moves that work off the tag but keeps the radio on longer, which also drains the battery. The team will settle this trade-off by measuring actual CSI capture rates, IMU data rates, and power draw on the prototype hardware.

The second challenge is the variability of indoor radio signals. The Wi-Fi measurements recorded at a given spot change from moment to moment as people walk through hallways, doors open and close, and network traffic varies [3], so the matching method must tolerate measurements that never exactly repeat the survey. The third challenge is IMU drift. Low-cost motion sensors accumulate position error within seconds [5], so Wi-Fi fixes must arrive often enough, and be reliable enough, to keep the EKF from diverging from the true position.

***Relevant Standards***

Four sets of standards constrain how BevoTag uses the radio spectrum, connects its sensors, and protects its data. Appendix A lists the specific clauses of each standard and how they affect the design.

IEEE 802.11 [6] defines how Wi-Fi devices format and exchange signals. It determines which Wi-Fi chips can report CSI, because CSI is derived from the training symbols that the standard places at the start of every packet, and it governs how the tag connects to the campus network.

The Federal Communications Commission (FCC) regulates radio emissions in the United States under Title 47 of the Code of Federal Regulations, Part 15 [7]. Part 15 limits both the intended Wi-Fi transmissions and the unintended electrical noise that the tag's processor and circuits emit, so that the tag does not interfere with campus communications. Choosing a Wi-Fi module that already holds FCC certification is the most practical way to meet these limits.

The IEEE 1451 family of standards [8] describes how sensors should present their data and calibration information to a processor. Following its approach keeps the IMU and Wi-Fi drivers sep-

arate from the position-estimation code, so that either sensor can be replaced without rewriting the filter.

Finally, NIST Special Publication 800-88 [9] and FIPS 140-3 [10] set federal guidelines for erasing stored data and for the cryptographic methods used to protect it. Because the tag stores network credentials and location history, these guidelines inform how it encrypts its reports and how a factory reset erases its memory.

***Ethical Considerations***

A device whose purpose is to report location creates ethical obligations that extend beyond the team's own users. The team has identified four areas of responsibility: the privacy of bystanders, the security of stored location data, the shared campus network, and the safety and environmental impact of the hardware.

The first obligation concerns people near the tag. To capture CSI, the tag listens to Wi-Fi traffic that may come from other people's devices. The firmware will therefore record only the signal measurements and packet header fields needed for positioning, such as the access point address and signal strength. It will not decode, store, or transmit the contents of anyone's network traffic.

The second obligation is to protect location data once collected. A history of where an item has been reveals where its owner has been, so the data are personal information. The team will encrypt data in transit between the tag and the server, restrict server access to authenticated owners, retain location history only as long as needed for the project, and ensure that a factory reset erases stored credentials and history, consistent with the guidelines in Appendix A.

The third obligation is to the campus network that BevoTag depends on. The tag will listen passively or connect as an ordinary client and will never transmit traffic designed to disrupt, congest, or probe UT Austin's network. It will use only networks, such as `utexas-iot`, that the university provides for devices of this kind [4].

The fourth obligation is the physical safety and environmental impact of the device. The tag will carry a small lithium battery near people and their belongings, so the design will include protection against overcharging, short circuits, and overheating. To limit electronic waste, the team will favor a rechargeable battery over disposable cells and will document how to dispose of the prototype's battery properly.

### *Project Deliverables*

At the end of the project, the team will deliver an integrated hardware and software prototype that estimates and displays the location of a tagged item within the surveyed campus areas. The prototype will consist of five deliverables:

- **BevoTag hardware device:** a compact device with a custom circuit board, enclosure, and battery that collects Wi-Fi and motion measurements and sends them to the localization system.
- **Embedded firmware:** software on the tag that reads the sensors, timestamps each measurement, monitors device status, and communicates over Wi-Fi.
- **Campus map:** a digital 3D map of the designated test areas that contains the spatial information needed to display and evaluate the tag's estimated location.
- **Localization backend:** server software that receives data from each tag, estimates its position from the Wi-Fi and motion measurements, and provides the result to the user application.
- **User application:** a web or mobile application in which a user can view a tag's estimated location and status on the campus map.

## Requirements Specification

A successful BevoTag implementation must provide continuous location estimates for a tagged item across representative indoor and outdoor areas of campus and must present them in a form that a user can readily interpret. The specifications in this subsection translate the design problem into measurable terms. They define the system's inputs and outputs, its user interface, the conditions under which it must operate, and the performance it must achieve, without prescribing a particular implementation. Wi-Fi and inertial sensing form the primary method; other positioning methods, such as GNSS outdoors, may be added if they improve performance.

### *Input and Output Specifications*

The BevoTag system consists of three components that must work together to produce a location estimate: the tag, a backend server, and the user application. Figure 2 shows the inputs each component receives, the outputs it produces, and the communication paths between them. The tag measures Wi-Fi signals, motion, and, when available outdoors, GNSS position, and it sends these measurements and its status to the server. The server estimates the tag's position and passes it to the application, which displays it to the user. A direct USB-C connection between the tag and a computer running the application allows configuration and debugging. These paths may change as the team makes further design decisions.

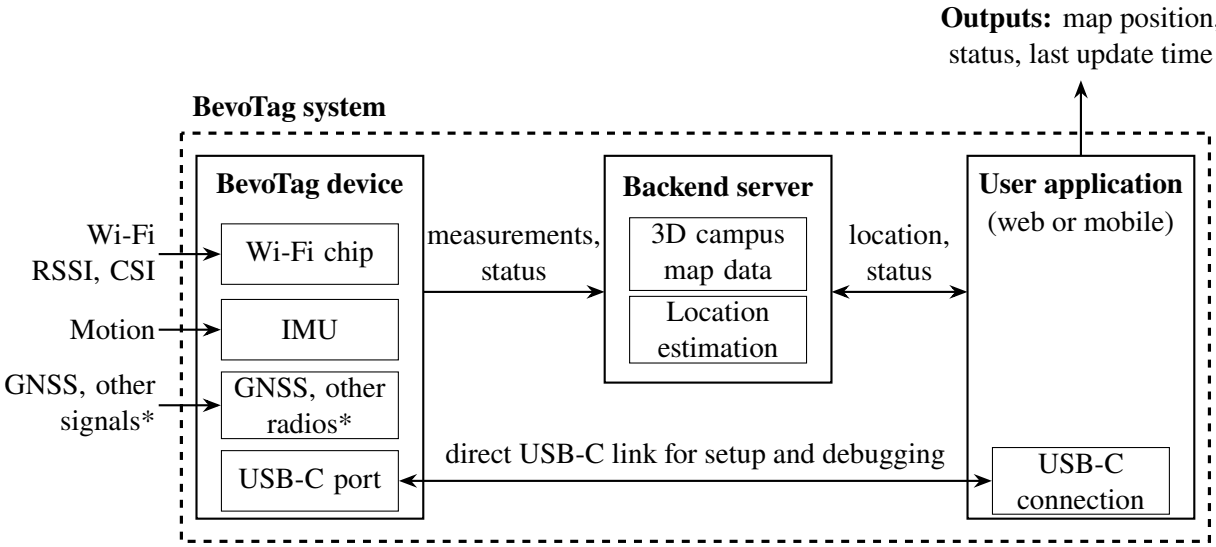



**Figure 2. BevoTag Input and Output Diagram (*optional)**

Table 1 lists each input and output in Figure 2, along with its expected values and units. The Wi-Fi measurements and motion data are the primary inputs; GNSS is used only when it is available and more accurate than Wi-Fi, such as in open outdoor areas.

**Table 1. BevoTag Input and Output Specifications**

| Input/Output | Description | Value or Range | Units |
|---|---|---|---|
| GNSS input | Satellite position, used outdoors when more accurate than Wi-Fi | Latitude, longitude, altitude | degrees, m |
| Wi-Fi CSI input | Channel state information from nearby Wi-Fi signals | Amplitude and phase per sub-channel | N/A |
| Wi-Fi RSSI input | Received strength of each nearby Wi-Fi signal | Device-dependent received power | dBm |
| IMU input | Acceleration and rotation measurements for dead reckoning | 3-axis acceleration, 3-axis angular rate | $m/s^2$, rad/s |
| Telemetry | Data exchanged between tag and backend | Measurements, location, status | N/A |
| Location output | Estimated position of the tag | Latitude, longitude, campus map coordinates | degrees, m |
| Timestamp output | Time of the latest location estimate | Date and time | date, h:min:s |
| Device status output | Current operating state of the tag | Battery, connectivity, sensor, processing state | N/A |
| Application output | Information presented to the user | Map position, status, last update time | N/A |

***User Interface Specifications***

The user application is the primary interface between the localization system and the user. It must allow a user to select a tag, view its estimated location on a campus map together with the time the tag was at that location, and check the tag's status. The application must also accept a direct USB-C connection to a tag for configuration and debugging. The initial implementation may be either a web or mobile application, provided that it supports the requirements in Table 2.

**Table 2. User Interface Specifications**

| Interface | Requirement | Value or Criterion |
|---|---|---|
| Device location | Show the tag's location on a 3D map of campus | Location displayed on map |
| Device selection | Let the user select or identify the tag to monitor | At least one tag |
| Location update | Update the displayed position when a new estimate arrives | Display matches latest estimate |
| Device status | Show connectivity and device status | Status shown for each tag |
| Map interaction | Let the user pan, zoom, and navigate the 3D map | Standard map controls for the platform |
| USB-C connection | Exchange data directly between a connected tag and the application | USB-C link between computer and tag |

***Operating Environment Specifications***

BevoTag must operate in representative indoor and outdoor areas of campus. Because the prototype will be tested within designated zones rather than across the entire campus, the operating environment is defined by the conditions expected in those zones: walls and floors that block signals, pedestrian traffic, changes in the tag's orientation, and access to UT Austin's wireless network. Table 3 summarizes these conditions.

***Performance Specifications***

Successful operation requires more than producing a location value; the estimates must be accurate, frequent, and timely enough for BevoTag to serve as a useful tracker. Existing Wi-Fi positioning systems show that meter-level accuracy is achievable, but performance varies widely with the sensing method and the radio environment [3]. The specifications in Table 4 therefore define performance in terms of repeatable measurements.

**Table 3. Operating Environment Specifications**

| Condition | Description | Value or Range | Units |
|---|---|---|---|
| Deployment area | Surveyed indoor and outdoor zones on campus, beginning with EER | Surveyed test zones | N/A |
| Radio environment | Signals may arrive directly (line of sight) or only after reflection or obstruction | Indoor and outdoor; line of sight and blocked | N/A |
| Wi-Fi band | Frequency band the tag must support | 2.4 | GHz |
| Network access | Test zones must be covered by the `utexas-iot` network [4] | IEEE 802.11 | N/A |
| Temperature | Ambient temperature range for correct operation | 0 to 40 | °C |
| User motion | Speed of the person carrying the tag | 0 to 5 | m/s |

The team will evaluate performance at surveyed points whose true positions are known independently of BevoTag's estimate. Tests will include both stationary and moving trials, indoors and outdoors, and will be repeated at different times of day to capture the effect of changing conditions such as pedestrian traffic.

**Table 4. Performance Specifications**

| Parameter | Description | Value or Range | Units |
|---|---|---|---|
| Median position error | Median horizontal distance between estimated and true position during trials | $< 5$ | m |
| Room identification accuracy | Fraction of trials in which the tag is placed in the correct surveyed room or region | $> 90$ | % |
| IMU sampling rate | Accelerometer and gyroscope sampling rate during active tracking | $\geq 100$ | Hz |
| End-to-end latency | Maximum time from a location estimate to its display in the application | $< 5$ | s |
| Time to first fix | Maximum time from startup to the first usable position estimate | $< 30$ | s |

## TEAM ORGANIZATION

The five-member BevoTag team is organized into two sub-teams, one for embedded hardware and firmware and one for localization and application software, coordinated by a project lead. All members contribute to system integration, testing, and documentation. This section first assigns roles according to each member's expertise and then sets the policies the team will follow

for communication, accountability, and decision-making.

**Project Roles**

The team divided its responsibilities according to the system's three major functions: collecting measurements, estimating position, and presenting results to the user. Each member owns specific development and verification tasks, and the two sub-teams will combine these contributions into a working prototype.

***Sub-Teams***

The Embedded Hardware and Firmware sub-team, Esteban Martinez and Li Shi, is responsible for the tag itself. It will first build a data logger for feasibility experiments and Wi-Fi survey collection, then integrate the required processing on the tag and evaluate measurement reliability, execution time, memory use, and power consumption.

The Localization and Application Software sub-team, Andrew Yin and Abhirit Das, is responsible for turning measurements into a location and presenting it to the user. Its work spans survey data processing, the position-estimation pipeline, the backend that receives and manages tag data, and the user application. The two sub-teams will work together to integrate the tag, the positioning software, and the application, and all members will take part in system testing.

***Personnel and Team-Wide Roles***

Ethan Chua brings positioning research experience from the UT Radionavigation Laboratory and project experience from Texas Aerial Robotics. As Project Lead, he will coordinate milestones, track progress, lead integration testing, and serve as the primary contact with Dr. Vikalo and the Technical TA.

Esteban Martinez and Li Shi have completed ECE 445L, Embedded Systems Design Lab, and have embedded systems experience through Texas Aerial Robotics and industry internships. Esteban will lead component selection, power-system design, hardware integration, and electrical testing. Li will lead firmware development, including Wi-Fi CSI and RSSI capture, IMU sampling, measurement timestamping, and network communication.

Andrew Yin and Abhirit Das have also completed ECE 445L. Andrew's experience in mobile application development and machine learning supports his work on Wi-Fi survey processing, position estimation, sensor fusion, and backend development. Abhirit brings experience in web application development and in profiling machine-learning deployment tools at NXP Semicon-

ductors, and he will develop and test the user application. Table 5 summarizes each member's assignment.

**Table 5. Project Responsibilities**

| Member | Sub-Team | Planned Contributions | Areas of Expertise | Related Coursework |
|---|---|---|---|---|
| Ethan Chua | Project Lead | Coordination, mentor communication, system integration, verification | RF positioning, GNSS, signal processing, sensor fusion, autonomous systems | ECE 351M Digital Signal Processing |
| Esteban Martinez | Embedded Hardware and Firmware | Component selection, power-system design, hardware integration, testing | Embedded hardware and firmware design | ECE 445L, ECE 422C, ECE 360C |
| Li Shi | Embedded Hardware and Firmware | Firmware, sensor drivers, data collection, Wi-Fi communication | Embedded hardware and firmware design | ECE 445L, ECE 422C, ECE 360C |
| Andrew Yin | Localization and Application Software | Survey processing, localization, sensor fusion, backend | Mobile development, software engineering, machine learning | ECE 445L, ECE 422C, ECE 360C |
| Abhirit Das | Localization and Application Software | User application, map display, status display, interface testing | Web development, software engineering, embedded systems | ECE 445L, ECE 422C, ECE 360C |

## Team Policies

With five members working in parallel on hardware, firmware, positioning algorithms, and the user application, the team needs shared rules for communicating, tracking work, holding members accountable, and resolving disagreements. The following policies apply to every member and sub-team, and the team will revisit them at the end of the semester as it prepares for detailed design.

### *Project Management Tools*

The team uses Discord as its primary communication channel, with separate channels for general coordination, each sub-team, and meeting logistics so that discussions stay organized and searchable. Members are expected to answer direct questions within 24 hours on weekdays and to flag time-sensitive issues, such as a blocked task or an approaching deadline, by mentioning the af-

fected members directly. The Project Lead coordinates email with Dr. Vikalo and the Technical TA and copies the full team so that every member receives the same guidance.

The team meets twice a week and meets with Dr. Vikalo every two weeks. A shared Google Calendar holds meeting times, course deadlines, and internal milestones. Firmware, application code, data processing scripts, and circuit board and enclosure designs are kept under version control in a shared GitHub repository, where GitHub Issues and a GitHub Projects board record each task's owner, due date, and status. Reports, meeting notes, and survey data are stored in a shared Google Drive folder.

***Team Expectations***

Each member is expected to attend both weekly team meetings and every mentor meeting, having reviewed the agenda beforehand, and to notify the team on Discord at least 24 hours in advance of an unavoidable absence. A member who misses a meeting is responsible for reading the notes and following up on any action items assigned to them. Internal deadlines fall at least 48 hours before each course deadline so the team has time to integrate, review, and revise its work.

Every deliverable should meet the standard the team would present to its mentor. Code is reviewed by at least one other member through a pull request before it is merged, hardware designs are checked by a second member before boards are ordered, and written sections are proofread by another member against the ECE Style Guide. At the start of each meeting, every member briefly reports what they completed, what they plan to do next, and any blockers, and the Project Lead records these updates and keeps the Projects board current. A member who expects to miss an internal deadline must tell the team at least 48 hours in advance so the work can be re-scoped or redistributed.

***Conflict Resolution***

The team makes significant decisions, such as selecting a positioning approach, choosing components, or changing the project scope, by majority vote after the options have been researched. The members most familiar with the question present the trade-offs of each option, supported by datasheets, published results, cost estimates, or the team's own measurements, and the outcome and its rationale are recorded in the meeting notes. Decisions that affect only one sub-team are made within that sub-team and reported at the next meeting.

Technical disagreements are settled with evidence rather than persistence. Where practical, the team runs a quick test or trade study against the performance specifications in Table 4 before

voting, and it brings any unresolved disagreement that poses significant risk to Dr. Vikalo or the Technical TA. Interpersonal conflicts are first addressed privately between the members involved; if that does not resolve the issue, an uninvolved member mediates, and the team escalates any conflict it cannot resolve to the Technical TA or the course instructor.

When a member does not meet a responsibility, the team responds in escalating steps. A first missed commitment is raised privately to understand the cause and agree on a recovery plan. Repeated missed deadlines or meetings are discussed at a team meeting, and corrective actions are recorded in the notes. If the problem continues, the team will notify the Technical TA and the course instructor and will report each member's contributions accurately in peer evaluations.

## CONCLUSION

This Project Definition documents the team's understanding of the BevoTag design problem and how the team has organized itself to solve it. Satellite positioning fails inside campus buildings, and existing trackers depend on closed networks of other people's phones, so the team proposes a low-cost tag that locates itself by matching campus Wi-Fi measurements against a building survey and fusing them with motion-sensor data. The report defined the functions, scope, standards, and ethical obligations of that design, set measurable requirements including a median error under 5 m and room identification above 90%, and assigned roles and working policies to the team.

The project carries real risk: CSI must be captured on low-power hardware, room-level accuracy depends on careful building surveys, and campus network coverage and access policies may limit where the tag can operate. Even so, the team's combined experience in embedded systems, positioning, and software gives it a realistic path to a working prototype within a set of surveyed buildings. Next, the team will revise this document using feedback from the Technical TA and the Writing Consultation, complete a data logger for initial Wi-Fi and IMU measurements, survey EER as the first test building, and refine the system block diagram and component selection.

## REFERENCES


[1] Apple Inc. (2026). *AirTag* [Online]. Available: https://www.apple.com/airtag/

[2] J. E. Yoder and T. E. Humphreys, "Low-cost inertial aiding for deep-urban tightly coupled multi-antenna precise GNSS," *NAVIGATION*, vol. 70, no. 1, art. no. navi.561, 2023, doi: 10.33012/navi.561.

[3] M. Jiang and J. Yuan, "A device free high-precision indoor positioning and tracking method based on GMM-WKNN algorithm," *Sci. Rep.*, vol. 15, no. 1, art. no. 17586, May 2025, doi: 10.1038/s41598-025-02876-1.

[4] The University of Texas at Austin. (2025, Aug. 25). *Connecting to utexas-iot Wi-Fi*, UT ServiceNow Knowledge Base KB0017215 [Online]. Available: https://ut.service-now.com/kb_view.do?sysparm_article=KB0017215

[5] M. Mukherjee *et al.*, "How IMU drift influences multi-radar inertial odometry for ground robots in subterranean terrains," arXiv:2602.24192, Feb. 2026.

[6] *IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*, IEEE Standard 802.11-2020, 2021.

[7] *Radio Frequency Devices*, 47 C.F.R. Part 15, 2025.

[8] *IEEE Standard for a Smart Transducer Interface for Sensors and Actuators—Common Functions, Communication Protocols, and Transducer Electronic Data Sheet (TEDS) Formats*, IEEE Standard 1451.0-2007, 2007.

[9] *Guidelines for Media Sanitization*, NIST Special Publication 800-88 Rev. 1, Dec. 2014.

[10] *Security Requirements for Cryptographic Modules*, FIPS PUB 140-3, Mar. 2019.

# APPENDIX A – APPLICABLE STANDARDS

This appendix lists the clauses of each standard introduced in the Problem Statement that most directly affect BevoTag's design. Each table gives the requirement and its consequence for the design, and the accompanying text explains how the team will verify compliance.

### IEEE 802.11: Wi-Fi Physical and MAC Layers

IEEE 802.11 [6] determines both how BevoTag can measure CSI and how it may join campus networks. Table 6 summarizes the relevant clauses. The team will verify CSI capture by comparing recorded sub-carrier data against known access-point beacons, confirm passive listening with a Wireshark capture showing no transmissions during scans, and confirm network association through connection logs on `utexas-iot`.

**Table 6. IEEE 802.11 Requirements Affecting BevoTag**

| Clause | Requirement | Effect on Design |
|---|---|---|
| Clauses 17 and 19 (OFDM and HT PHY) | Define the sub-carrier structure and training fields at the start of each packet | The Wi-Fi chip must expose per-sub-carrier CSI from these training fields for 20 and 40 MHz channels |
| Clause 10 (MAC sub-layer) | Defines scanning, frame headers, and medium access | Firmware reads only header fields (access point address, signal strength) and listens passively without probing |
| 802.11i/802.1X security | Defines WPA2/WPA3 association and authentication | Telemetry uses the university's supported authentication on `utexas-iot` |
| Channel plan | Defines 2.4 GHz channel frequencies and bandwidths | The radio uses standard non-overlapping channels (1, 6, 11) |

### FCC 47 CFR Part 15: Radio Frequency Devices

Part 15 [7] limits both intentional transmissions and unintended emissions from the tag's digital circuits. Table 7 summarizes the relevant sections. Because the tag will use a module that already holds an FCC modular grant, the team will verify conducted output power on a bench spectrum analyzer and check the board for unintended emissions with a near-field probe, rather than performing full certification testing.

**Table 7. FCC Part 15 Requirements Affecting BevoTag**

| Section | Requirement | Effect on Design |
|---|---|---|
| §15.109 (unintentional radiators) | Radiated emission limits for Class B digital devices | Processor clocks, sensor buses, and power conversion need careful layout and decoupling |
| §15.247 (intentional radiators) | 2400–2483.5 MHz output power limit of 30 dBm (36 dBm EIRP) | Use a pre-certified Wi-Fi module transmitting at about 12–20 dBm |
| §15.205 and §15.209 (restricted bands) | Limits on harmonics and spurious emissions | Rely on the certified module's filtering and follow its antenna layout rules |
| §15.19 and §15.21 (labeling) | Compliance statement and warning against modification | Enclosure and documentation carry the Part 15 statement and module FCC ID |

### IEEE 1451: Smart Transducer Interface

IEEE 1451 [8] guides how the firmware separates sensor hardware from the processing that uses its data. Table 8 summarizes the relevant parts.

**Table 8. IEEE 1451 Guidelines Affecting BevoTag**

| Part | Guideline | Effect on Design |
|---|---|---|
| 1451.0 (common functions) | Common data model and transducer electronic data sheets (TEDS) | A hardware abstraction layer separates sensor drivers from the EKF |
| 1451.2 (digital interface) | Sensor-to-processor interfaces such as SPI, $I^2C$, and UART | IMU connected by SPI (up to 10 MHz) or $I^2C$ fast mode (400 kHz) |
| 1451.4 (virtual TEDS) | Self-describing calibration data | IMU scale factors and bias offsets stored in a standard format in flash |
| 1451.5 (wireless) | Wireless transfer of sensor data | Telemetry packets carry IMU, Wi-Fi, and battery data in a defined format |

The team will adopt the architecture of IEEE 1451 rather than its full command set, which would use more memory than the tag can spare. Compliance will be checked by firmware review, logic-analyzer timing of the sensor bus, and unit tests confirming that calibration data load correctly at startup.

**NIST SP 800-88 and FIPS 140-3: Data Sanitization and Cryptography**

These federal guidelines [9], [10] govern how the tag protects and erases the credentials and location history it stores. Table 9 summarizes the relevant sections. The team will verify sanitization by reading back flash memory through a debug probe after a factory reset, verify encryption against published NIST test vectors, and verify secure boot by attempting to load an unsigned firmware image.

**Table 9. Data Security Requirements Affecting BevoTag**

| Section | Requirement | Effect on Design |
|---|---|---|
| SP 800-88 “Clear” | Overwrite stored data so that simple recovery fails | Factory reset overwrites stored credentials, location history, and survey tables |
| SP 800-88 cryptographic erase | Destroy encryption keys so stored data cannot be decrypted | Reset erases the keys held in the chip’s secure storage |
| FIPS 140-3 Level 1 | Use approved algorithms such as AES and SHA | Telemetry encrypted with AES-GCM using the chip’s hardware accelerator |
| FIPS 140-3 roles and authentication | Separate administrative functions from normal use | Secure boot accepts only signed firmware images |